§2.3

P.1. $\psi_2(t)-\psi_1(t)=e^{2t}$

$\psi_3(t)-\psi_2(t)=1+e^{2t}$.

are two linearly independent solutions of the homogeneous equation. Therefore,

$$y(t)=\tilde{c}_1e^{2t}+\tilde{c}_2(1+e^{2t})+t^2$$
$$=\tilde{c}_2+(\tilde{c}_1+\tilde{c}_2)e^{2t}+t^2=c_1+c_2e^{2t}+t^2$$

P3. Solution:

$$y_1(t)=\psi_2(t)-\psi_1(t)$$
$$=4e^t$$
$$y_2(t)=\psi_3(t)-\psi_2(t)$$
$$=-2e^t+e^{-t^3}$$

are two linearly independent solution of the homogeneous equation $L[y]=0$. Thus,

$$y(t)=\tilde{c}_1y_1(t)+\tilde{c}_2y_2(t)+\psi_1(t)$$
$$=4\tilde{c}_1e^t+\tilde{c}_2(-2e^t+e^{-t^3})$$
$$+3e^t+e^{t^2}$$
$$=(4\tilde{c}_1-2\tilde{c}_2+3)e^t+\tilde{c}_2e^{-t^3}+e^{t^2}$$
$$=c_1e^t+c_2e^{-t^3}+e^{t^2}$$

is the general solution of $L[y]=g(t)$.

$$\left.\begin{aligned}y(0)=1&\Rightarrow c_1+c_2+1=1\\ y'(0)=2&\Rightarrow c_1=2\end{aligned}\right\}\Rightarrow \begin{aligned}c_1&=2\\ c_2&=-2\end{aligned}$$

$$\Rightarrow y(t)=2e^t-2e^{-t^3}+e^{t^2}$$

P4. Solution:

Let $\psi_1(t)$ and $\psi_2(t)$ be any two solution of the equation.

$L[y]=ay''+by'+cy=g(t)$.

i.e., $L[\psi_1]=g(t)$

$L[\psi_2]=g(t)$

$\Rightarrow L[\psi_1-\psi_2]=0$, and characteristic equation is.

$$ar^2+br+c=0$$
$$r=\frac{-b\pm\sqrt{b^2-4ac}}{2a}.$$

(i) $b^2-4ac>0$,

Let $\alpha=-\frac{b}{2a}$, $\beta=\sqrt{b^2-4ac}/2a$.

then.

$$(\psi_1-\psi_2)(t)=c_1e^{(\alpha+\beta)t}+c_2e^{(\alpha-\beta)t}$$
$$\lim_{t\to+\infty}(\psi_1-\psi_2)(t)=\lim_{t\to+\infty}c_1e^{(\alpha+\beta)t}+c_2e^{(\alpha-\beta)t}$$
$$=0$$

Since $\alpha+\beta=\frac{-b+\sqrt{b^2-4ac}}{2a}$

$$<\frac{-b+b}{2a}=0$$
$$\alpha-\beta=\frac{-b-\sqrt{b^2-4ac}}{2a}<0$$

(ii) $b^2-4ac<0$.

Let $\alpha=-\frac{b}{2a}$, $\beta=\sqrt{4ac-b^2}/2a$

then

$$(\psi_1-\psi_2)(t)=e^{\alpha t}(c_1\cos\beta t+c_2\sin\beta t)$$
$$\lim_{t\to+\infty}|(\psi_1-\psi_2)(t)|=\lim_{t\to\infty}|e^{\alpha t}|\,|c_1\cos\beta t+c_2\sin\beta t|$$
$$\le\lim_{t\to\infty}|e^{\alpha t}|\cdot(|c_1|+|c_2|)$$
$$=0\qquad(\alpha<0)$$

(iii) $b^2-4ac=0$.

let $\alpha=-\frac{b}{2a}$, clearly $\alpha<0$

$$(\psi_1-\psi_2)(t)=[c_1+c_2t]e^{-\alpha t}$$
$$\lim_{t\to+\infty}(\psi_1-\psi_2)(t)=\lim_{t\to+\infty}c_1e^{-\alpha t}+\lim_{t\to+\infty}\frac{c_2t}{e^{\alpha t}}$$
$$=0$$